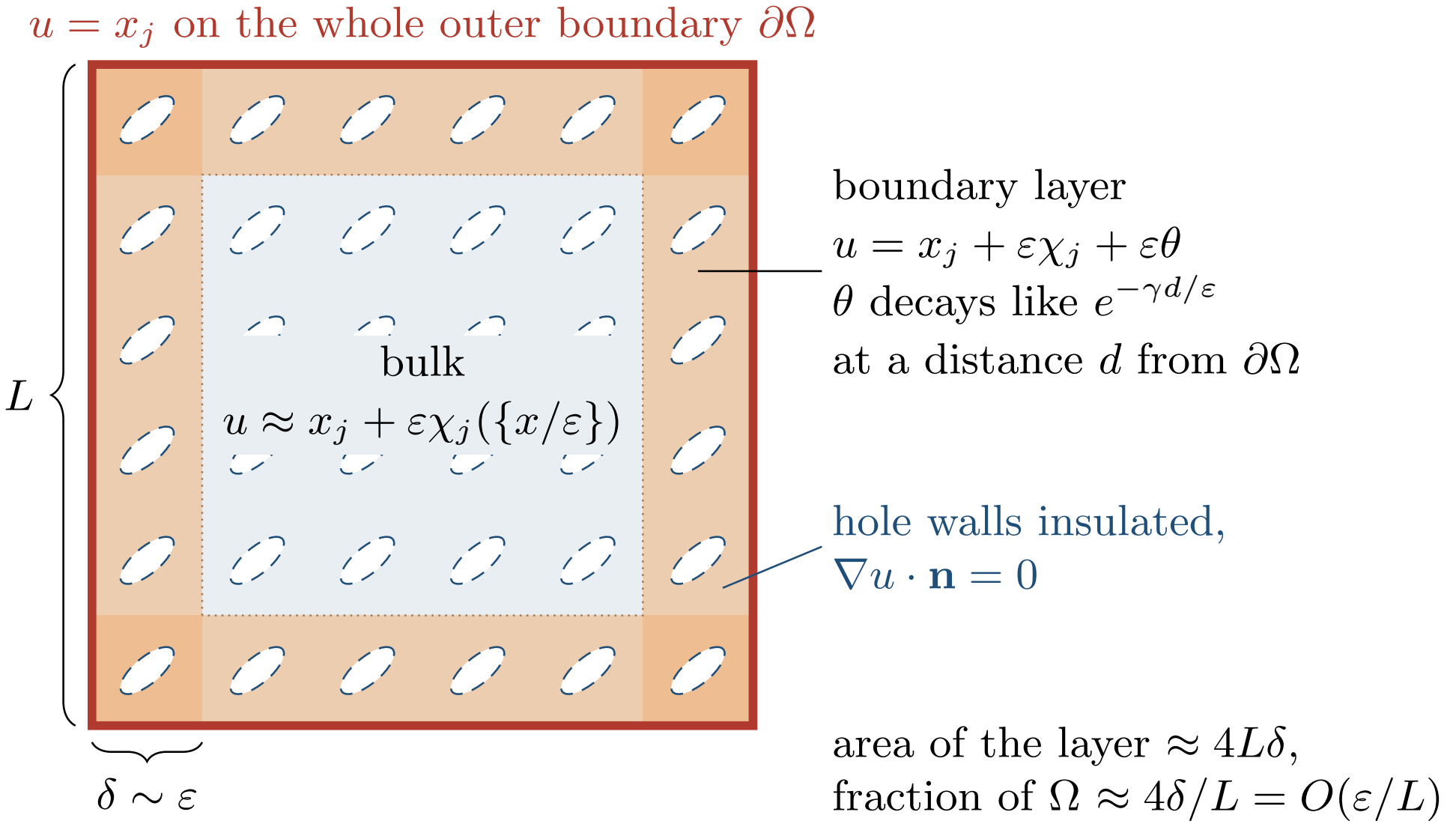

$u = x_j$ on the whole outer boundary $\partial\Omega$
L
$\delta \sim \varepsilon$
bulk
$u \approx x_j + \varepsilon\chi_j(\{x/\varepsilon\})$
boundary layer
$u = x_j + \varepsilon\chi_j + \varepsilon\theta$
θ decays like $e^{-\gamma d/\varepsilon}$
at a distance d from $\partial\Omega$
hole walls insulated,
$\nabla u \cdot \mathbf{n} = 0$
area of the layer $\approx 4L\delta$,
fraction of $\Omega \approx 4\delta/L = O(\varepsilon/L)$